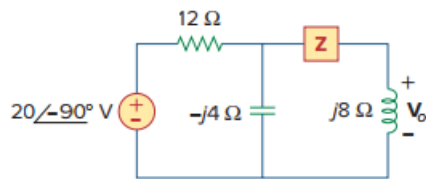


Problem

Find $\mathbf{Z}$ in the network of Fig. 9.62, given that $\mathbf{V}_o = 4\angle 0^\circ \text{ V}$.

Figure 9.62:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram in Figure 9.62 in the textbook.

Draw the circuit with node notations.

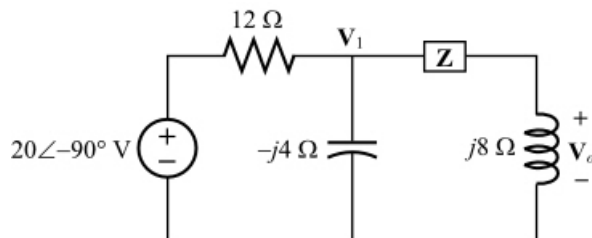


Figure 1

Step 2 of 3

Apply Kirchhoff's current law at node $\mathbf{V}_1$.

$$\frac{\mathbf{V}_1 - 20\angle -90^\circ}{12} + \frac{\mathbf{V}_1}{-j4} + \frac{\mathbf{V}_1}{\mathbf{Z} + j8} = 0$$

$$\frac{\mathbf{V}_1}{12} + \frac{\mathbf{V}_1}{-j4} + \frac{\mathbf{V}_1}{\mathbf{Z} + j8} = \frac{20\angle -90^\circ}{12}$$

$$\left(\frac{1}{12} + \frac{1}{-j4} + \frac{1}{\mathbf{Z} + j8} \right) \mathbf{V}_1 = \frac{20\angle -90^\circ}{12}$$

$$\left(\frac{(-j4)(\mathbf{Z} + j8) + (12)(\mathbf{Z} + j8) + (12)(-j4)}{(12)(-j4)(\mathbf{Z} + j8)} \right) \mathbf{V}_1 = \frac{20\angle -90^\circ}{12}$$

$$\begin{aligned} \mathbf{V}_1 &= \frac{(20\angle -90^\circ)(-j4)(\mathbf{Z} + j8)}{(-j4)(\mathbf{Z} + j8) + (12)(\mathbf{Z} + j8) + (12)(-j4)} \\ &= \frac{-80(\mathbf{Z} + j8)}{-j4\mathbf{Z} + 32 + 12\mathbf{Z} + j96 - j48} \\ &= \frac{-80(\mathbf{Z} + j8)}{(12 - j4)\mathbf{Z} + 32 + j48} \end{aligned}$$

Step 3 of 3

The output voltage is,

$$\mathbf{V}_o = \left(\frac{j8}{\mathbf{Z} + j8} \right) \mathbf{V}_i$$

Substitute $4\angle 0^\circ \text{ V}$ for $\mathbf{V}_o$ and $\frac{-80(\mathbf{Z} + j8)}{(12 - j4)\mathbf{Z} + 32 + j48}$ for $\mathbf{V}_i$ in the equation.

$$4\angle 0^\circ \text{ V} = \left(\frac{j8}{\mathbf{Z} + j8} \right) \left(\frac{-80(\mathbf{Z} + j8)}{(12 - j4)\mathbf{Z} + 32 + j48} \right)$$

$$4\angle 0^\circ \text{ V} = \frac{-j640}{(12 - j4)\mathbf{Z} + 32 + j48}$$

$$(12 - j4)\mathbf{Z} + 32 + j48 = \frac{-j640}{4\angle 0^\circ \text{ V}}$$

$$(12 - j4)\mathbf{Z} = -j160 - 32 - j48$$

$$\mathbf{Z} = \frac{-32 - j208}{12 - j4}$$

$$= 2.8 - j16.4 \, \Omega$$

Therefore, the value of impedance, $\mathbf{Z}$ is $\boxed{2.8 - j16.4 \, \Omega}$.